

Topics in Error-Correcting Codes

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Lecture 3

Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.

Logistics

All the logistics for this class can be found on the [course website](#).

Recap: q -ary Entropy and Bounds

Recall the Hamming and Gilbert–Varshamov bounds and their asymptotic forms from last lecture.

Definition 1. For integers $n, \ell \geq 0$, let the volume of a Hamming ball of radius ℓ in Σ^n , where $|\Sigma| = q$, be denoted by $\text{Vol}_q(n, \ell)$. Then

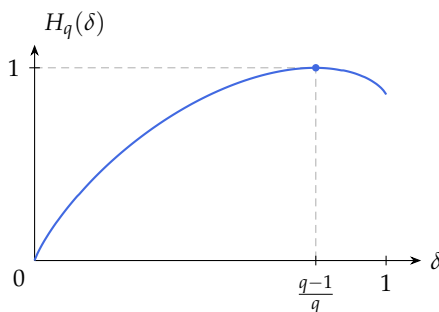
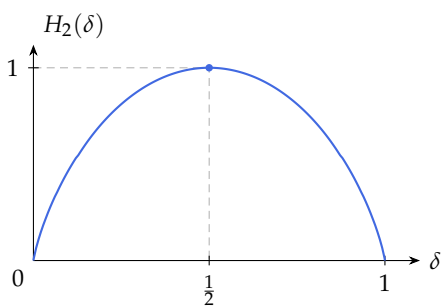
$$\text{Vol}_q(n, \ell) = \sum_{j=0}^{\ell} \binom{n}{j} (q-1)^j.$$

Theorem 2 (Gilbert-Varshamov bound). For any $1 \leq d \leq n$, there exists a q -ary code $\mathcal{C}$ of length n and distance at least d with

$$|\mathcal{C}| \geq \frac{q^n}{\text{Vol}_q(n, d-1)}.$$

Definition 3. For $q \geq 2$ and $\delta \in (0, 1)$, the q -ary entropy function is

$$H_q(\delta) = \delta \log_q(q-1) - \delta \log_q(\delta) - (1-\delta) \log_q(1-\delta).$$



Proposition 4: Let $q \geq 2$ be an integer and $0 < \delta \leq 1 - 1/q$. Then:

- $\text{Vol}_q(n, \lfloor \delta n \rfloor) \leq q^{H_q(\delta)n}$;
- $\text{Vol}_q(n, \lfloor \delta n \rfloor) \geq q^{(H_q(\delta) - o(1))n}$.

Theorem 5 (Asymptotic Hamming). Let $q \geq 2$ be fixed, and let $\mathcal{C} = \{\mathcal{C}_i\}_{i \in \mathbb{N}}$ be a family of q -ary codes. Then

$$R(\mathcal{C}) \leq 1 - H_q(\delta(\mathcal{C})/2).$$

Theorem 6 (Asymptotic GV). Let $q \geq 2$ be fixed. For every $\delta \in (0, 1 - 1/q)$, there exists a family of q -ary codes $\mathcal{C} = \{\mathcal{C}_i\}_{i \in \mathbb{N}}$ such that $\delta(\mathcal{C}) \geq \delta$ and

$$R(\mathcal{C}) \geq 1 - H_q(\delta).$$

Open Question 7. Can you construct a family of binary codes meeting the GV bound in deterministic polynomial time with positive rate and distance?

Theorem 8 (Random linear codes achieve the GV bound). Let q be a prime power, let $0 < \delta < 1 - 1/q$, and let $0 < \varepsilon < 1 - H_q(\delta)$. For sufficiently large n , set

$$k = \lfloor (1 - H_q(\delta) - \varepsilon)n \rfloor$$

and choose G uniformly at random from $\mathbb{F}_q^{k \times n}$. If $\mathcal{C} = \{mG \mid m \in \mathbb{F}_q^k\}$, then

$$\Pr_G[\mathcal{C} \text{ is an } (n, k, \lceil \delta n \rceil)_q \text{ linear code}] \geq 1 - q^{-\varepsilon n}.$$

Proof. Fix a nonzero message $m \in \mathbb{F}_q^k$. Since G is uniformly random, mG is uniformly distributed over $\mathbb{F}_q^n$. Hence

$$\Pr_G[\text{wt}(mG) < \lceil \delta n \rceil] = \frac{\text{Vol}_q(n, \lceil \delta n \rceil - 1)}{q^n} \leq q^{-(1 - H_q(\delta))n}.$$

Taking a union bound over all nonzero $m \in \mathbb{F}_q^k$ gives

$$\Pr_G[\exists m \neq 0 : \text{wt}(mG) < \lceil \delta n \rceil] \leq q^k q^{-(1 - H_q(\delta))n} \leq q^{-\varepsilon n}.$$

Outside this event, every nonzero message maps to a word of weight at least $\lceil \delta n \rceil$. In particular, the map $m \mapsto mG$ is injective, so G has rank k . The distance bound now follows from the fact that the minimum distance of a linear code is its minimum nonzero Hamming weight, since $\Delta(c, c') = \text{wt}(c - c')$. $\square$

Question 9. What happens in the regime when $\delta \geq 1 - 1/q$?

Plotkin Bound

The bound of Plotkin [1960] addresses the question above: over a fixed alphabet, positive asymptotic rate is impossible when the relative distance is at least $1 - 1/q$.

We use the formulation in Guruswami–Rudra–Sudan, Theorem 4.4.1, with the equality case sharpened to qn .

Theorem 10 (Plotkin Bound). *Let $\mathcal{C} \subseteq \Sigma^n$ be an $(n, k, d)_q$ code, where $|\Sigma| = q$. Then:*

1. *If $d = (1 - 1/q)n$, then $|\mathcal{C}| \leq qn$.*

2. *If $d > (1 - 1/q)n$, then*

$$|\mathcal{C}| \leq \frac{qd}{qd - (q - 1)n}.$$

Proof. We first prove part 2. Write $M = |\mathcal{C}|$ and sum the distances over all ordered pairs of codewords. Since distinct codewords have distance at least d ,

$$M(M - 1)d \leq \sum_{c, c' \in \mathcal{C}} \Delta(c, c').$$

For each coordinate $i \in [n]$ and symbol $a \in \Sigma$, let $N_{i,a} = |\{c \in \mathcal{C} \mid c_i = a\}|$. We have $\sum_{a \in \Sigma} N_{i,a} = M$, so Cauchy–Schwarz gives

$$\sum_{a \in \Sigma} N_{i,a}^2 \geq \frac{M^2}{q}.$$

Counting the pairs that disagree in each coordinate, we obtain

$$\begin{aligned} \sum_{c, c' \in \mathcal{C}} \Delta(c, c') &= \sum_{i=1}^n \left(M^2 - \sum_{a \in \Sigma} N_{i,a}^2 \right) \\ &\leq n \left(1 - \frac{1}{q} \right) M^2. \end{aligned}$$

Combining these inequalities and dividing by M yields

$$\left(d - \left(1 - \frac{1}{q} \right) n \right) M \leq d.$$

When $d > (1 - 1/q)n$, the coefficient of M is positive. Dividing by it proves part 2.

For part 1, suppose $d = (1 - 1/q)n$. At least M/q codewords share the same symbol in their first coordinate. Keep those codewords and delete that coordinate, obtaining a code $\mathcal{C}'$ of length $n - 1$ and pairwise distances at least d . Since $d > (1 - 1/q)(n - 1)$, the bound just proved gives

$$\frac{M}{q} \leq |\mathcal{C}'| \leq \frac{qd}{qd - (q - 1)(n - 1)} = \frac{qd}{q - 1} = n.$$

Thus $M \leq qn$, proving part 1. □

Corollary 11 (Asymptotic Plotkin Bound). *Let $q \geq 2$ be fixed, and let $\mathcal{C} = \{C_i\}_{i \in \mathbb{N}}$ be a family of q -ary codes. Then*

$$R(\mathcal{C}) \leq \max \left\{ 0, 1 - \frac{q}{q-1} \delta(\mathcal{C}) \right\}.$$

In particular, if $\delta(\mathcal{C}) \geq 1 - 1/q$, then $R(\mathcal{C}) = 0$.

Proof. For an $(n, k, d)_q$ code of size $M = q^k$, let

$$m = \min \left\{ n, \left\lceil \frac{qd}{q-1} \right\rceil - 1 \right\}, \quad \ell = n - m.$$

Keep the codewords with the most frequent pattern on the first ℓ coordinates, then delete those coordinates. The resulting code $\mathcal{C}'$ has length m , pairwise distances at least $d > (1 - 1/q)m$, and size at least M/q^ℓ . By finite Plotkin,

$$\frac{M}{q^\ell} \leq |\mathcal{C}'| \leq \frac{qd}{qd - (q-1)m} \leq qn,$$

since the denominator is a positive integer. Thus

$$\frac{k}{n} \leq \frac{\ell + \log_q(qn)}{n} \leq \max \left\{ 0, 1 - \frac{qd}{(q-1)n} \right\} + O\left(\frac{\log n}{n}\right).$$

Apply this to the family and use $d_i/n_i \geq \delta(\mathcal{C}) - o(1)$. Taking limits proves the claim. $\square$

Remark (Plotkin versus Singleton). Recall that the asymptotic Singleton bound is $R \leq 1 - \delta$. For every fixed alphabet size q and $0 < \delta \leq 1 - 1/q$, Plotkin gives the strictly stronger bound

$$R \leq 1 - \frac{q}{q-1} \delta = 1 - \delta - \frac{\delta}{q-1} < 1 - \delta.$$

For $1 - 1/q < \delta < 1$, Plotkin instead gives $R = 0 < 1 - \delta$. Thus no family with positive rate and positive relative distance can attain the Singleton bound over a fixed alphabet. This obstruction holds for every fixed q . To reach the Singleton line with both rate and relative distance positive, the alphabet must grow with the block length.

Dual Codes

We recall the definition introduced at the end of last lecture.

Definition 12 (Dual Code). Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a linear code of dimension k with parity-check matrix $H \in \mathbb{F}_q^{n \times (n-k)}$, so that $\mathcal{C} = \{c \in \mathbb{F}_q^n \mid cH = 0\}$. The *dual code* $\mathcal{C}^\perp$ is the code generated by H^T . Equivalently,

$$\mathcal{C}^\perp = \{x \in \mathbb{F}_q^n \mid x \cdot c = 0 \text{ for all } c \in \mathcal{C}\},$$

where $x \cdot c = \sum_{i=1}^n x_i c_i$ is computed over $\mathbb{F}_q$.

Remark. The dual code has dimension $n - k$, and $(\mathcal{C}^\perp)^\perp = \mathcal{C}$. H^T is a generator matrix for $\mathcal{C}^\perp$, while G^T is a parity-check matrix for $\mathcal{C}^\perp$ whenever G generates $\mathcal{C}$.

Hadamard Codes

Definition 13 (Hadamard Code). For an integer $r \geq 2$, index the coordinates by the nonzero vectors $x \in \mathbb{F}_2^r$. The *Hadamard code* is

$$\text{Had}_r = \{(a \cdot x)_{x \in \mathbb{F}_2^r \setminus \{0\}} \mid a \in \mathbb{F}_2^r\} \subseteq \mathbb{F}_2^{2^r-1},$$

where the dot products are computed over $\mathbb{F}_2$.

Remark. Our definition differs from the standard length- 2^r Hadamard code by omitting the coordinate indexed by $x = 0$, which is always zero.

Proposition 14 (Parameters of the Hadamard Code): *The code Had_r is a binary linear code with parameters*

$$(2^r - 1, r, 2^{r-1})_2.$$

Proof. The encoding is linear from the r -dimensional message space $\mathbb{F}_2^r$.

For $a \neq 0$, the map $x \mapsto a \cdot x$ has a kernel of dimension $r - 1$, so the encoded word has weight exactly $2^r - 2^{r-1} = 2^{r-1}$. $\square$

Lemma 15. *For every $r \geq 2$, the dual code $\text{Had}_r^\perp$ has parameters*

$$(2^r - 1, 2^r - r - 1, 3)_2,$$

the same parameters claimed for Hamming codes in the previous lecture.

Proof. The length is $n = 2^r - 1$, and duality gives dimension $n - r = 2^r - r - 1$. It remains to show that the minimum distance is exactly 3.

Let $G \in \mathbb{F}_2^{r \times n}$ have as its columns all the nonzero vectors of $\mathbb{F}_2^r$, in the order used to index the coordinates. Then G generates Had_r , so G^T is a parity-check matrix for its dual. For a word $c = (c_x)_{x \neq 0}$, this means

$$c \in \text{Had}_r^\perp \iff cG^T = 0 \iff \sum_{x \neq 0} c_x x = 0.$$

A word of weight 1 cannot satisfy this condition, since every column is nonzero. A word of weight 2 cannot satisfy it either: two distinct nonzero vectors cannot sum to zero over $\mathbb{F}_2$. Hence the minimum distance is at least 3.

Since $r \geq 2$, choose distinct nonzero vectors $u, v \in \mathbb{F}_2^r$. The vectors $u, v, u + v$ are distinct and nonzero, and

$$u + v + (u + v) = 0.$$

Thus the word with ones precisely in those three coordinates belongs to $\text{Had}_r^\perp$ and has weight 3. This proves equality. $\square$

Remark. In fact, $\text{Had}_r^\perp$ is the Hamming code itself, up to the ordering of coordinates.

Recap: Singleton Bound

Recall the Singleton bound.

Theorem 16 (Singleton Bound). *For every $(n, k, d)_q$ code $\mathcal{C} \subseteq \Sigma^n$, where $|\Sigma| = q$, we have*

$$k + d \leq n + 1.$$

Definition 17 (MDS Code). A code meeting the Singleton bound with equality, so that $d = n - k + 1$, is called a *maximum distance separable* (MDS) code.

Reed–Solomon Codes

What happens if we allow the alphabet to grow? A basic fact about polynomials will let us attain the Singleton bound exactly!

Fact 18 (Degree Mantra). A nonzero univariate polynomial of degree at most t over a field has at most t distinct roots!

We now give the construction of [Reed and Solomon \[1960\]](#).

Definition 19 (Reed–Solomon Code). Let q be a prime power, let $1 \leq k \leq n \leq q$, and choose distinct evaluation points $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$. The *Reed–Solomon code* with these evaluation points and message length k is

$$\text{RS}_{q,k}(\alpha_1, \dots, \alpha_n) = \{(f(\alpha_1), \dots, f(\alpha_n)) \mid f \in \mathbb{F}_q[X], \deg f < k\},$$

including the zero polynomial.

To encode a message $m = (m_0, \dots, m_{k-1}) \in \mathbb{F}_q^k$, regard its entries as the coefficients of the polynomial

$$f_m(X) = \sum_{j=0}^{k-1} m_j X^j$$

and output its evaluations:

$$\text{Enc}(m) = \text{Enc}(f_m) = (f_m(\alpha_1), \dots, f_m(\alpha_n)).$$

Theorem 20 (Reed–Solomon Codes Meet Singleton). *For every prime power q and $1 \leq k \leq n \leq q$, the Reed–Solomon construction gives a linear $(n, k, n - k + 1)_q$ code. In particular, it is MDS.*

Proof. The code $\mathcal{C}$ is linear. For every nonzero polynomial f of degree less than k , the degree mantra gives $\text{wt}(\text{Enc}(f)) \geq n - k + 1$. Therefore,

$$\Delta(\mathcal{C}) = \min_{0 \neq c \in \mathcal{C}} \text{wt}(c) \geq n - k + 1.$$

Equality is attained by $f(X) = \prod_{j=1}^{k-1} (X - \alpha_j)$, whose encoding has exactly $k - 1$ zero coordinates. $\square$

Reed–Solomon codes meet the Singleton bound exactly and are MDS, with a growing alphabet ($q \geq n$), as needed.

References

- Morris Plotkin. Binary codes with specified minimum distance. *IRE Transactions on Information Theory*, 6(4):445–450, September 1960. doi: 10.1109/TIT.1960.1057584.
- Irving S. Reed and Gustave Solomon. Polynomial codes over certain finite fields. *Journal of the Society for Industrial and Applied Mathematics*, 8(2):300–304, June 1960. doi: 10.1137/0108018.